Table 2: MM-GBSA binding free energy decomposition for HPV16 E6/E7-derived wild-type (WT) and altered peptide ligand (APL) complexes with HLA-A*02:01.

Peptide	$\Delta E_{\mathbf{vdW}}$ **(kcal/mol)**	$\Delta E_{\mathbf{elec}}$ **(kcal/mol)**	$\Delta G_{\mathbf{polar}}$ **(kcal/mol)**	$\Delta G_{\mathbf{nonpolar}}$ **(kcal/mol)**	$\Delta G_{\mathbf{bind}}$ **(kcal/mol)**
E6wt (10-mer)	-86.74 ± 6.71	-185.56 ± 21.78	201.77 ± 20.52	-13.60 ± 0.66	-84.14 ± 8.80
E6apl (10-mer)	-82.44 ± 5.53	-270.22 ± 26.73	291.97 ± 22.16	-12.91 ± 0.49	-73.59 ± 8.83
E7wt1 (9-mer)	-83.87 ± 4.91	-257.49 ± 36.12	278.36 ± 29.71	-12.46 ± 0.39	-75.47 ± 9.20
E7apl1 (9-mer)	-90.46 ± 4.40	-240.53 ± 27.31	263.77 ± 23.09	-12.99 ± 0.40	-80.21 ± 7.12
E7wt2 (10-mer)	-79.43 ± 6.65	-263.58 ± 26.96	277.50 ± 22.48	-12.59 ± 0.55	-78.09 ± 9.25
E7apl2 (10-mer)	-80.21 ± 5.95	-299.54 ± 38.91	310.46 ± 34.03	-12.95 ± 0.54	-82.23 ± 9.53

Note: **WT**, wild-type; **APL**, altered peptide ligand. Energy terms: $\Delta E_{\mathbf{vdW}}$, van der Waals energy; $\Delta E_{\mathbf{elec}}$, electrostatic energy; $\Delta G_{\mathbf{polar}}$, polar solvation (generalized Born); $\Delta G_{\mathbf{nonpolar}}$, nonpolar solvation (surface area); $\Delta G_{\mathbf{bind}}$, total predicted binding free energy.

Values are mean $\pm$ standard deviation of per-frame MM-GBSA estimates. Calculations were performed using the single-trajectory approach on exactly 600 frames extracted at 10-frame intervals from the 20–50 ns equilibrium window of the 50 ns trajectory. The Onufriev GB-OBC II model ($igb = 5$) was utilized with a physiological salt concentration of 0.15 M. Conformational entropy ($-T\Delta S$) was not included. Peptide nomenclature follows Table 1. Because the per-frame standard deviations are large relative to the WT–APL differences, all six complexes appear energetically favorable and stable. Therefore, the MM-GBSA estimates should be interpreted primarily as indicators of complex stability rather than precise predictors of relative binding affinity.